

Vuttaravilli Ambedkar

Embedded Software / Validation Engineer

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PROFESSIONAL SUMMARY:

Embedded Software and Validation Engineer with 4 years of experience in embedded systems and protocol validation. Strong knowledge in C programming and communication protocols such as PCIe, NVMe, RS422, RS485, and MIL-STD-1553. Currently working on pre-silicon validation using Protium and Palladium platforms, focusing on PCIe link behavior, ASPM L1, PMD3 power states, and NVMe command flows. Good at debugging, log analysis, and system-level understanding.

TECHNICAL SKILLS:

- Programming: C, Embedded C
- Protocols: PCIe, NVMe(Admin and IO command flow , NSID mapping basics), AXI, RS422, RS485, UART, basic knowledge on SPI and I2C, MIL-STD-1553
- Embedded Systems: Interrupts, DMA (basic), AXI protocol basics (read/write channels, handshake, burst types, ordering), free RTOS (basic).
- Validation & Debug: Pre-Silicon Validation, Log Analysis, Debugging.
- Tools: Xilinx SDK, VS Code, Keil, MPLABx
- Platforms: Protium, Palladium
- Operating Systems: Linux (vim editor, basic commands, log analysis), Windows
- Other: Git, Code Review, AI-tools (ChatGPT, Copilot, Claude).

PROFFESIONAL EXPERIENCE:

AMD

Role: Validation / Embedded Engineer

Duration: May 2025 – Present

- Performed pre-silicon validation of PCIe and NVMe subsystems using Protium and Palladium
- Analyzed PCIe link training and link rate behavior
- Worked on ASPM L1 and observed power state transitions
- Debugged PMD3 (D3 state) related issues
- Performed log analysis and basic failure debugging
- Observed and validated NVMe command flow (submission and completion queues)
- Worked on NVMe firmware concepts including Admin and IO command handling and NSID-based mapping (basic understanding)
- Analyzed AXI protocol behavior at signal level, including read/write transactions, VALID/READY handshake , burst type backpressure, and ordering
- Worked with team members to debug system-level issues
- Used VS Code, vim, and AI tools for development and debugging

DRDO (RCI / DSP)

Role: Embedded Software Engineer

Duration: April 2022 – April 2025

Project: Onboard Mission Software

- Worked on embedded drivers for communication systems
- Worked on RS422 communication
- Worked on MIL-STD-1553 protocol integration and validation
- Worked with RTOS-based system requirement for 6 months
- Supported debugging and testing activities

Project: Launch Pad Computer Software

- Worked on RS422 communication module for system checkout
- Built GUI using Q Creator for real-time monitoring
- Validated communication between subsystems

Project: Bus Management Unit (BMU)

- Worked on RS485 and RS422 communication interfaces and interacted with few devices like Rate sensor and Reaction wheel
- Worked on MIL-STD-1553 BRM register-level configuration
- Interfaced communication between FPGA and embedded system
- Debugged communication-related issues

EDUCATION

- B. Tech – Electronics and Communication Engineering
- JNTU Kakinada | 2014 – 2018 | 68.74%

PROJECT HIGHLIGHTS

- Experience in aerospace and mission-critical systems
- Exposure to PCIe and NVMe protocols
- Experience in debugging and validation experiment in pre-silicon environment

(V. AMBEDKAR)